King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Final Exam Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. $\int \dfrac{5x-2}{(x-2)^2}\,dx =$

(a) $-2\ln|x-2| + \dfrac{2/5}{(x-2)^2} + c$

(b) $5\ln|x-2| - 2\ln|x+2| + c$

(c) $\ln|x-2| - \ln|5x-2| + c$

(d) $5\ln|x-2| - \dfrac{8}{x-2} + c$

(e) $\dfrac{1}{2}\ln|x-2| + \dfrac{5}{(x-2)^2} + c$

2. The sum of the series $\sum_{n=0}^{\infty}\left(\dfrac{3}{2^n} - \dfrac{2}{3^n}\right)$ is

(a) 0

(b) 6

(c) 4

(d) 3

(e) 2

3. If f is continuous on the interval $[0,b]$ where $f(x)+f(b-x)\neq 0$ on $[0,b]$, then the value of $\int_0^b \dfrac{f(x)}{f(x)+f(b-x)}\,dx =$

(a) $\dfrac{1}{2}b$

(b) $2b$

(c) $2b+\sqrt{2}$

(d) $2b+1$

(e) $\sqrt{2b}$

4. The area of the region between the curves $f(x) = -2^x$ and $g(x) = 1 - 3x$ is equal to

(a) $4 - \dfrac{3}{\ln 2}$

(b) $10 - \dfrac{6}{\ln 2}$

(c) $4 + \dfrac{3}{\ln 2}$

(d) $8 + \dfrac{1}{\ln 2}$

(e) $6 + \dfrac{1}{\ln 2}$

5. For positive values of p, the series $\sum_{n=1}^{\infty} \left(\dfrac{3}{2p}\right)^n$ converges if

(a) $0 < p \leq \dfrac{3}{2}$

(b) $p > 0$

(c) $p \geq \dfrac{3}{2}$

(d) $p > \dfrac{3}{2}$

(e) $0 < p < \dfrac{3}{2}$

6. The volume of the solid whose base is bounded by the circle $x^2 + y^2 = 9$ and whose cross sections perpendicular to the x-axis are equilateral triangles is equal to

(a) 36

(b) $36\sqrt{3}$

(c) $18\sqrt{3}$

(d) 72

(e) $9\sqrt{3}$

7. If L is the length of the curve $f(x) = \cosh x$ over $[0, 2]$, then

(a) $4 < L < 5$

(b) $2 < L < 3$

(c) $5 < L < 6$

(d) $3 < L < 4$

(e) $1 < L < 2$

8. If $F(x) = \int_x^{x+2} (4t+1)\,dt$, then $F'(x) =$

(a) 2

(b) 10

(c) 4

(d) 8

(e) 6

9. $\int_0^{\pi/2} \dfrac{1}{3 - 2\cos\theta}\,d\theta =$

(a) $\dfrac{\sqrt{5}}{5}\tan^{-1}\left(\dfrac{\sqrt{5}}{2}\right)$

(b) $\dfrac{1}{2}\tan^{-1}\left(\dfrac{\sqrt{5}}{5}\right)$

(c) $\dfrac{2}{\sqrt{5}}\tan^{-1}(\sqrt{5})$

(d) $\dfrac{\sqrt{5}}{5}\tan^{-1}\left(\dfrac{1}{\sqrt{5}}\right)$

(e) $\tan^{-1}\left(\dfrac{\sqrt{5}}{5}\right)$

10. $\int_1^3 \dfrac{e^{3/x}}{x^2}\,dx =$

(a) $\dfrac{1}{3}e^3 - 1$

(b) $\dfrac{e^3 + e}{\sqrt{3}}$

(c) $\dfrac{e^3 + e}{2}$

(d) $\dfrac{e^3 - e}{\sqrt{3}}$

(e) $\dfrac{1}{3}(e^3 - e)$

11. $\int_0^\pi \sin^5 x\, dx =$

(a) $\frac{16}{15}$

(b) $\frac{19}{15}$

(c) $\frac{13}{15}$

(d) $\frac{11}{15}$

(e) $\frac{17}{15}$

12. If $p_2(x)$ is the second Maclaurin polynomial for the function $f(x) = \cos \pi x$, then $p_2\left(\frac{1}{\pi}\right) =$

(a) 0

(b) 1

(c) $\frac{1}{\pi}$

(d) $\frac{1}{4}$

(e) $\frac{1}{2}$

13. The series $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$ converges

(a) only for $x \in [-1, 1)$

(b) only for $x \in [0, 1]$

(c) only for $x \in [-1, 1]$

(d) only at $x = 0$

(e) for all x

14. Which of the following series is/are divergent?

$$\text{I. } \sum_{n=1}^{\infty} \frac{5}{n!} \qquad \text{II. } \sum_{n=1}^{\infty} \frac{9^n}{n^5} \qquad \text{III. } \sum_{n=2}^{\infty} e^{-3n}$$

(a) II only

(b) II and I

(c) I only

(d) II and III

(e) I and III

15. The infinite series $\sum_{n=1}^{\infty} \dfrac{1}{n(n+2)}$ converges to

(a) 1

(b) $\dfrac{1}{4}$

(c) 0

(d) $\dfrac{1}{2}$

(e) $\dfrac{3}{4}$

16. $\int_e^{\infty} \dfrac{1}{x \ln x}\, dx =$

(a) $e\sqrt{e}$

(b) e

(c) $e\sqrt{e} - 1$

(d) ∞

(e) $\dfrac{e}{\sqrt{e}}$

17. The volume of the solid generated by revolving the region bounded by the graphs of $y = 2x - x^2$ and $y = 0$ about the line $x = 4$ is equal to:

(a) 8π

(b) 6π

(c) 4π

(d) $\dfrac{8}{3}\pi$

(e) $\dfrac{4}{3}\pi$

18. If $f(x) = \sum_{n=2}^{\infty} \left(\dfrac{x}{2}\right)^n$, then the interval of convergence of $\int f(x)\, dx$ is

(a) $(-2, 2)$

(b) $[-2, 2]$

(c) $(-2, 2]$

(d) $[-2, 2)$

(e) $(-\infty, \infty)$

19. The series $\sum_{n=1}^{\infty} \dfrac{\arctan n}{n^2+1}$

(a) converges to $\dfrac{\pi^2}{32}$

(b) diverges because $\lim_{n\to\infty} \dfrac{\arctan n}{n^2+1} \neq 0$

(c) converges to $\dfrac{3\pi^2}{32}$

(d) diverges because $\int_1^{\infty} \dfrac{\arctan x}{x^2+1}\,dx = \infty$

(e) converges

20. The least number of terms required to approximate the sum of the series $\sum_{n=0}^{\infty} \dfrac{(-1)^n}{2n^3-1}$ with an error of less than 0.01 is

(a) 5

(b) 6

(c) 4

(d) 3

(e) 2

21. The sequence whose nth term $a_n = \dfrac{2^n}{3^n-1}$ converges to

(a) $\ln 3$

(b) $\ln \dfrac{1}{3}$

(c) 1

(d) 0

(e) $\ln 2$

22. $\int \sec^3 x\,dx =$

(a) $\dfrac{\sqrt{3}}{2}|\sec x \tan x| + \dfrac{1}{2}\ln|\sec x + \tan x| + c$

(b) $\dfrac{1}{2}\tan x + \sec x \ln|\sec x + \tan x| + c$

(c) $2\sec x + \dfrac{\sqrt{3}}{2}\ln|\sec x + \tan x| + c$

(d) $2|\sec x \tan x| + \ln|\sec x + \tan x| + c$

(e) $\dfrac{1}{2}(\sec x \tan x + \ln|\sec x + \tan x|) + c$

23. The Power Series representation of $f(x) = \cos\sqrt{x}$ is

(a) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{2n+2}}{(2n)!}$

(b) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{n+1}}{(2n)!}$

(c) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{2n}}{n!}$

(d) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{n}}{(2n)!}$

(e) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{n}}{n!}$

24. The Power Series representation for the function $f(x) = \dfrac{5}{2x-3}$, centered at $c = -3$ is

(a) $\sum_{n=0}^{\infty} \dfrac{2^n}{9^n}(x-3)^n, \ \left|\dfrac{2}{9}(x+3)\right| < 1$

(b) $\sum_{n=0}^{\infty} -5\dfrac{2^n}{9^n}(x-3)^n, \ \left|\dfrac{2}{9}(x+3)\right| < 1$

(c) $-5\sum_{n=0}^{\infty} \dfrac{2^n}{9^{n+1}}(x+3)^n, \ \left|\dfrac{2}{9}(x+3)\right| < 1$

(d) $\sum_{n=0}^{\infty} 5\left(\dfrac{2}{9}\right)^n (x+3)^n, \ \left|\dfrac{2}{9}(x+3)\right| < 1$

(e) $\sum_{n=0}^{\infty} \dfrac{2^n}{9^{n+1}}(x-3)^n, \ \left|\dfrac{2}{9}(x+3)\right| < 1$

25. Which of the following series is/are convergent?

$$\text{I. } \sum_{n=0}^{\infty} \frac{4^n}{5^n+3} \qquad \text{II. } \sum_{n=2}^{\infty} \frac{\ln n}{n+1} \qquad \text{III. } \sum_{n=1}^{\infty} \frac{n}{n^2+1}$$

(a) I and II

(b) I, II and III

(c) I only

(d) I and III

(e) II and III

26. $\int_0^\infty \dfrac{4}{\sqrt{x}\,(x+6)}\,dx =$

(a) $\dfrac{4\pi}{\sqrt{6}}$

(b) $\dfrac{\pi}{\sqrt{6}} - 1$

(c) $\dfrac{\pi}{4\sqrt{6}}$

(d) $4\sqrt{6}$

(e) $\dfrac{2}{3}\pi - 1$

27. $\int \dfrac{e^x}{(e^{2x}+1)(e^x-1)}\,dx =$

(a) $\dfrac{1}{2}e^x\left(\ln|e^x-1| + \ln|e^{2x}-1|\right) + c$

(b) $\dfrac{1}{4}\ln\left|e^x - \dfrac{1}{2}\right| - \ln|e^{2x}+1| + x\tan^{-1}(e^x) + c$

(c) $\dfrac{1}{2}e^x\left(\ln|e^x+1| + \ln|e^{2x}+1| + \tan^{-1}(e^x-1)\right) + c$

(d) $\dfrac{1}{2}\ln|e^x-1| + \ln|e^{2x}-1| - x\tan^{-1}(e^x) + c$

(e) $\dfrac{1}{2}\left(\ln|e^x-1| - \dfrac{1}{2}\ln(e^{2x}+1) - \tan^{-1}(e^x)\right) + c$

28. Which of the following series is/are divergent?

I. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+1}$ II. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n\sqrt{n}}{\sqrt[3]{n}}$ III. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{e^n}$

(a) II only

(b) I and III

(c) II and I

(d) I only

(e) II and III

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$\frac{5x-2}{(x-2)^2} = \frac{5}{x-2} + \frac{8}{(x-2)^2}$.
2	**(d)**	$\frac{3}{1-1/2} - \frac{2}{1-1/3} = 6 - 3 = 3$.
3	**(a)**	$I = \int_0^b \frac{f(b-x)}{f(x)+f(b-x)}dx$, so $2I = b$, $I = \frac{b}{2}$.
4	**(b)**	Intersections $x = 1, 3$: $\int_1^3 (3x - 1 - 2^x)dx = 10 - \frac{6}{\ln 2}$.
5	**(d)**	Geometric, $r = \frac{3}{2p} < 1 \iff p > \frac{3}{2}$.
6	**(b)**	$A = \sqrt{3}(9 - x^2)$, $V = \int_{-3}^{3} A\,dx = 36\sqrt{3}$.
7	**(d)**	$L = \int_0^2 \cosh x\,dx = \sinh 2 \approx 3.63$.
8	**(d)**	$F' = (4(x + 2) + 1) - (4x + 1) = 8$.
9	**(c)**	$t = \tan\frac{\theta}{2}$: $\int_0^1 \frac{2\,dt}{1+5t^2} = \frac{2}{\sqrt{5}} \tan^{-1}\sqrt{5}$.
10	**(e)**	$u = 3/x$: $\frac{1}{3}\int_1^3 e^u\,du = \frac{1}{3}(e^3 - e)$.
11	**(a)**	$\int_{-1}^{1}(1 - u^2)^2 du = \frac{16}{15}$.
12	**(e)**	$p_2(x) = 1 - \frac{\pi^2 x^2}{2}$, so $p_2(1/\pi) = \frac{1}{2}$.
13	**(e)**	Ratio $= \frac{x^2}{(2n+2)(2n+1)} \to 0$: converges for all x.
14	**(a)**	Ratio test: I gives 0, II gives 9, III geometric $r = e^{-3}$.

Q	Ans	Brief explanation
15	**(e)**	Telescoping: $S_N \to \frac{1}{2}\left(1+\frac{1}{2}\right) = \frac{3}{4}$.
16	**(d)**	$\int_e^b = \ln(\ln b) \to \infty$: diverges.
17	**(a)**	Shells: $2\pi \int_0^2 (4-x)(2x-x^2)dx = 8\pi$.
18	**(d)**	Integrated series: $x=2$ diverges (harmonic), $x=-2$ converges (AST): $[-2,2)$.
19	**(e)**	$\leq \frac{\pi}{2n^2}$, a convergent p-series: converges.
20	**(d)**	$\lvert R_N\rvert \leq b_{N+1}$; $b_4 = \frac{1}{127} < 0.01$, so $N=3$.
21	**(d)**	$a_n = \frac{(2/3)^n}{1-3^{-n}} \to 0$.
22	**(e)**	By parts: $2I = \sec x \tan x + \ln\lvert\sec x + \tan x\rvert$.
23	**(d)**	$\cos u$ series with $u=\sqrt{x}$: $sum(-1)^n \frac{x^n}{(2n)!}$.
24	**(c)**	$\frac{5}{2x-3} = -\frac{5}{9}\frac{1}{1-\frac{2}{9}(x+3)} = -5\sum \frac{2^n}{9^{n+1}}(x+3)^n$.
25	**(c)**	I converges (geometric bound); II and III diverge (comparison with $1/n$).
26	**(a)**	$u=\sqrt{x}$: $\int_0^\infty \frac{8\,du}{u^2+6} = \frac{4\pi}{\sqrt{6}}$.
27	**(e)**	$u=e^x$: $\frac{1}{2}\ln\lvert u-1\rvert - \frac{1}{4}\ln(u^2+1) - \frac{1}{2}\tan^{-1}u$.
28	**(a)**	I converges (AST), II diverges ($n^{1/6}\to\infty$), III geometric converges.